

Quiz 6

NAME: _____

Q.1. $f(x) = 2x^2 + bx + c$ is some quadratic equation with unknown coefficients b, c . Its vertex is $(-1, -1)$. Find the coefficients b and c . (**Hint:** start by setting up the vertex form equation)

The vertex form of a quadratic polynomial, $f(x) = ax^2 + bx + c$, is $a(x - h)^2 + k$, where $V(h, k)$ is the vertex of the polynomial's graph $y = f(x)$.

In this problem, we can see that $a = 2$ and we are told that $V(h, k) = (-1, -1)$, so $h = -1, k = -1$. Plugging these values into the vertex form, we get that:

$$\begin{aligned} f(x) &= 2(x - (-1))^2 + (-1) \\ \Rightarrow f(x) &= 2(x + 1)^2 - 1 \\ \Rightarrow f(x) &= 2(x^2 + 2x + 1) - 1 \\ &= 2x^2 + 4x + 1 \end{aligned}$$

Comparing this with the form $2x^2 + bx + c$ (and equating coefficients of like terms), we see that $b = 4$ and $c = 1$.

Q.2. Find all x -intercepts of the circle with center $(4, 2)$ and radius 2. (**Hint:** start by setting up the equation of the circle)

The equation of a circle with center, $C(h, k)$, and radius, r , is:

$$(x - h)^2 + (y - k)^2 = r^2$$

In this problem, we have $C(h, k) = (4, 2)$ and $r = 2$, so our equation is:

$$\begin{aligned} (x - 4)^2 + (y - 2)^2 &= 2^2 \\ \Rightarrow (x - 4)^2 + (y - 2)^2 &= 4 \end{aligned}$$

We need to find the x -intercepts of the circle, the points where the circle intersects the x -axis. Remember that x -intercepts of a graph are exactly the points $(x, 0)$ on the graph (i.e. points on the graph where $y = 0$). So setting $y = 0$ in the above circle equation:

$$\begin{aligned} (x - 4)^2 + (0 - 2)^2 &= 4 \\ \Rightarrow (x - 4)^2 + \cancel{4} &= \cancel{4} \\ \Rightarrow (x - 4)^2 &= 0 \\ \Rightarrow x - 4 &= 0 \\ \Rightarrow x &= 4 \end{aligned}$$

so our circle has only one x -intercept, $(4, 0)$.

Q.3. Suppose we have a pair of functions $f(x) = x^2 - 2x + 1$ and $g(x) = x - 1$. First, state the domain of the function $\frac{f(x)}{g(x)}$ and then simplify the rational expression $\frac{f(x)}{g(x)}$.

Our function is:

$$\frac{f(x)}{g(x)} = \frac{x^2 - 2x + 1}{x - 1}$$

What are the possible domain restrictions for this kind of function? Well, we see a fraction, and we know that the denominator of a fraction cannot be 0, so that's a clear domain restriction. Specifically $x - 1 \neq 0 \Rightarrow x \neq 1$. But aside from that, nothing in the function indicates a potential domain issue, since there's no other denominator and no radicals.

Assuming we are in this domain, $x \neq 1$, then we may simplify this rational expression by factoring the numerator:

$$\begin{aligned} \frac{f(x)}{g(x)} &= \frac{x^2 - 2x + 1}{x - 1} \\ &= \frac{(x - 1)^2}{x - 1} \\ &= x - 1 \end{aligned}$$

So when simplified (over its domain), we have $\frac{f(x)}{g(x)} = x - 1$.

Q.4. Suppose we have a pair of functions $f(x) = \sqrt{x}$ and $g(x) = 2x + 4$. Compute the composition $(f \circ g)(x)$ and state the domain of this composite function in interval notation.

Remember how compositions work. They are two-step processes. In order to find $(f \circ g)(x)$, we plug x into g , getting a first output $g(x)$, and THEN plug $g(x)$ into f , getting a second, final output of $f(g(x))$:

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(2x + 4) \\ &= \sqrt{2x + 4} \end{aligned}$$

This function only has a domain restriction arising from the radical (remember that even-order roots are only well-defined for nonnegative values). This means our domain restriction is just $2x + 4 \geq 0 \Rightarrow x \geq -2$. So our function $(f \circ g)(x)$ is $\sqrt{2x + 4}$ and it has domain $[-2, \infty)$.